

ADNI Knowledge Graph Graph Classification Report

From Labeled Property Graph to Knowledge Graph:
Phase 1 Schema Migration Results

Project: ADNI Knowledge Graph for Alzheimer's Disease Research

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Phase: 1 (Schema Migration) — Steps 17–20 Complete

Table of Contents

1. Executive Summary
2. Definitions: LPG vs Knowledge Graph
3. ADNI Graph Before Phase 1 (LPG State)
4. Phase 1 Transformation Pipeline
5. ADNI Graph After Phase 1 (KG State)
6. Ontology Layer Detail
7. Side-by-Side Comparison: LPG vs KG
8. Classification Verdict
9. What This Enables for Phases 2-4

1. Executive Summary

On February 24, 2026, Phase 1 of the ADNI Knowledge Graph project was completed. This phase transformed the existing Neo4j Labeled Property Graph (LPG) — containing ~407,000 nodes and ~1.16 million relationships ingested from 108 ADNI clinical tables — into a semantically grounded **Knowledge Graph (KG)**.

The transformation was achieved **in-place**, without rebuilding the database, through four pipeline steps (17–20) that added:

- **12 uniqueness constraints** and **15 performance indexes** (Step 17)
- **Ontology codes** on all observation nodes: SNOMED-CT, LOINC, UBERON, ICD-10, MONDO, NCI Thesaurus (Step 18)
- **URI properties** on 30 relationship types covering ~1.2M relationships (Step 18)
- **52 OntologyConcept nodes** with IS_A hierarchies across 4 ontologies + ICD-10 (Steps 19-20)
- **126,743 new semantic edges**: 100,770 MAPS_TO + 25,946 CLASSIFIED_AS + 27 IS_A (Steps 19-20)

The graph now carries formal semantic meaning through ontology grounding, enabling machine reasoning, cross-ontology queries, and interoperability with external biomedical knowledge bases such as AlzKB. **By every standard definition, the graph is now a Knowledge Graph.**

2. Definitions: Labeled Property Graph vs Knowledge Graph

2.1 Labeled Property Graph (LPG)

A Labeled Property Graph is formally defined as a tuple $G = (V, E, \rho, \lambda, \sigma)$ where V is a set of vertices (nodes), E is a set of directed edges (relationships), ρ is a property function mapping nodes/edges to key-value pairs, λ is a labeling function assigning type identifiers, and σ represents optional schema constraints.

LPGs are a **database model** designed for efficient storage and traversal. They emphasize flexible schema evolution, fast multi-hop queries via index-free adjacency, and rich property storage on both nodes and edges. Critically, LPGs do **not** require global identifiers (URIs/IRIs) and have **no inherent semantic grounding** to external ontologies or formal type systems. Node labels and relationship types are application-specific strings.

2.2 Knowledge Graph (KG)

A Knowledge Graph is a knowledge base that uses a graph-structured data model to formally represent semantics by describing entities and their interrelationships. The critical distinction from an LPG is that a KG encodes **meaning** for programmatic use through ontologies, which describe entity types, characteristics, and relationships using standards such as:

- **OWL** (Web Ontology Language) — for logical inference and class hierarchies
- **RDFS** (RDF Schema) — for vocabulary definition (`rdfs:subClassOf = IS_A`)
- **SKOS** (Simple Knowledge Organization System) — for concept mapping (`skos:exactMatch = MAPS_TO`)
- **RO** (Relation Ontology) — for standardized biological relationship predicates
- **SNOMED-CT, LOINC, UBERON, ICD-10** — domain-specific ontology standards

A graph becomes a KG when its nodes and edges are grounded in formal ontologies, carry globally unique identifiers (URIs), participate in `IS_A` taxonomic hierarchies, and support machine reasoning through these formal semantics. The key phrase is: *a KG carries meaning that machines can interpret without application-specific knowledge.*

2.3 Can a Neo4j Graph Be Both?

Yes. Neo4j remains an LPG database at the infrastructure level (using its native property graph storage engine and Cypher query language), but by adding semantic properties (ontology codes, URIs, `rdf_type` annotations, `OntologyConcept` nodes, `IS_A/MAPS_TO` relationships), it is upgraded to a Knowledge Graph at the semantic layer. Recent literature describes this as a **Semantic Property Graph (SPG)** or **LPG-based Knowledge Graph**, combining RDF reasoning capabilities with property graph performance.

3. ADNI Graph Before Phase 1 (LPG State)

Before Phase 1, the ADNI graph was a pure Labeled Property Graph constructed through pipeline steps 1–16. It contained ~407,000 nodes across 17 label types and ~1.16 million relationships across 20+ types. All data from 108 ADNI CSV tables (~5,800 columns) was ingested, including patient demographics, cognitive assessments, biomarkers, imaging metadata, genetic profiles, and family history.

While comprehensive in coverage, the graph lacked any formal semantic grounding. Node labels like 'Diagnosis' and relationship types like 'HAS_VISIT' were application-specific strings with no connection to external ontologies. A 'diagnosis_code' of 'AD' was meaningful to humans but opaque to machines without custom application logic.

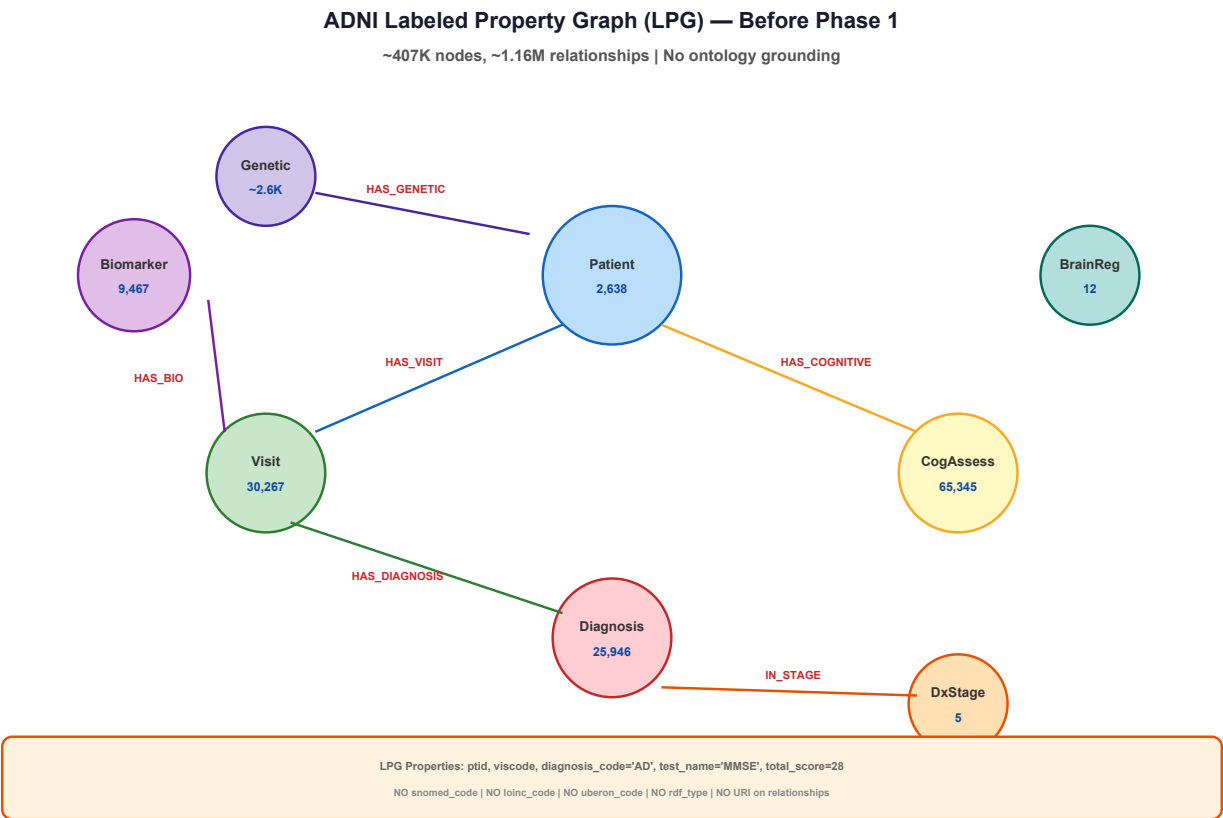


Figure 1: ADNI Labeled Property Graph before Phase 1. No ontology codes, no URI properties, no `OntologyConcept` nodes.

4. Phase 1 Transformation Pipeline

The transformation was executed through four sequential steps, each building on the previous:

- **Step 17: Apply Composite Unique Constraints** — Created 12 uniqueness constraints (5 core + 7 composite for observation nodes) and 15 performance indexes. All use IF NOT EXISTS for idempotency. Verified Neo4j 5.24.2 supports composite constraints.
- **Step 18: Add Ontology Properties (In-Place Upgrade)** — Enriched existing nodes with ontology codes without creating new nodes: Diagnosis nodes received snomed_code, icd10_code, and mondo_code; CognitiveAssessment nodes received loinc_code; Biomarker nodes received loinc_code; BrainRegion nodes received uberon_code; Patient and Visit nodes received rdf_type. 30 relationship types received URI properties. 100% coverage achieved.
- **Step 19: ICD-10 Integration** — Created 5 OntologyConcept nodes for ICD-10 codes (G30.9, G30, F06.7, F06, Z03.89), built 2 IS_A hierarchy edges, and linked all 25,946 Diagnosis nodes via CLASSIFIED_AS edges. WHO ICD REST API client implemented with static JSON fallback.
- **Step 20: Ontology Layer + MAPS_TO** — Created 47 OntologyConcept nodes across 4 ontologies (18 SNOMED-CT, 10 LOINC, 14 UBERON, 5 HPO). Built 25 IS_A hierarchy edges. Created 100,770 MAPS_TO edges linking data nodes to their ontology concepts.

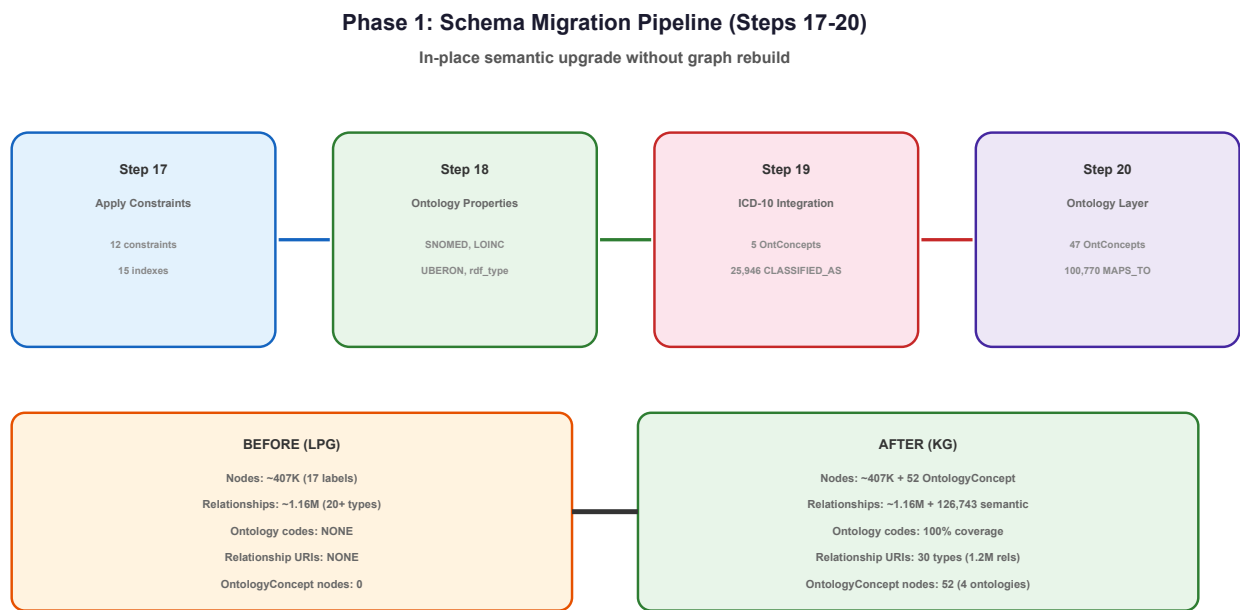


Figure 2: Phase 1 transformation pipeline showing the four sequential steps and before/after comparison.

5. ADNI Graph After Phase 1 (KG State)

After Phase 1, the graph carries full semantic meaning. Every Diagnosis node is grounded in SNOMED-CT (e.g., Alzheimer's disease = 26929004) and classified under ICD-10 (G30.9). Every CognitiveAssessment is linked to its LOINC code (e.g., MMSE = 72106-8). Every BrainRegion carries its UBERON identifier (e.g., Hippocampus = UBERON:0002421). All relationships carry URI properties from the Relation Ontology (e.g., HAS_VISIT = ro:RO_0000056).

The OntologyConcept layer provides formal IS_A taxonomies. A machine can now traverse from a specific Diagnosis node through MAPS_TO to a SNOMED-CT concept, then up the IS_A hierarchy: Alzheimer's disease IS_A Dementia IS_A Neurodegenerative disorder IS_A Disorder of nervous system IS_A Disease. This was impossible in the LPG state.

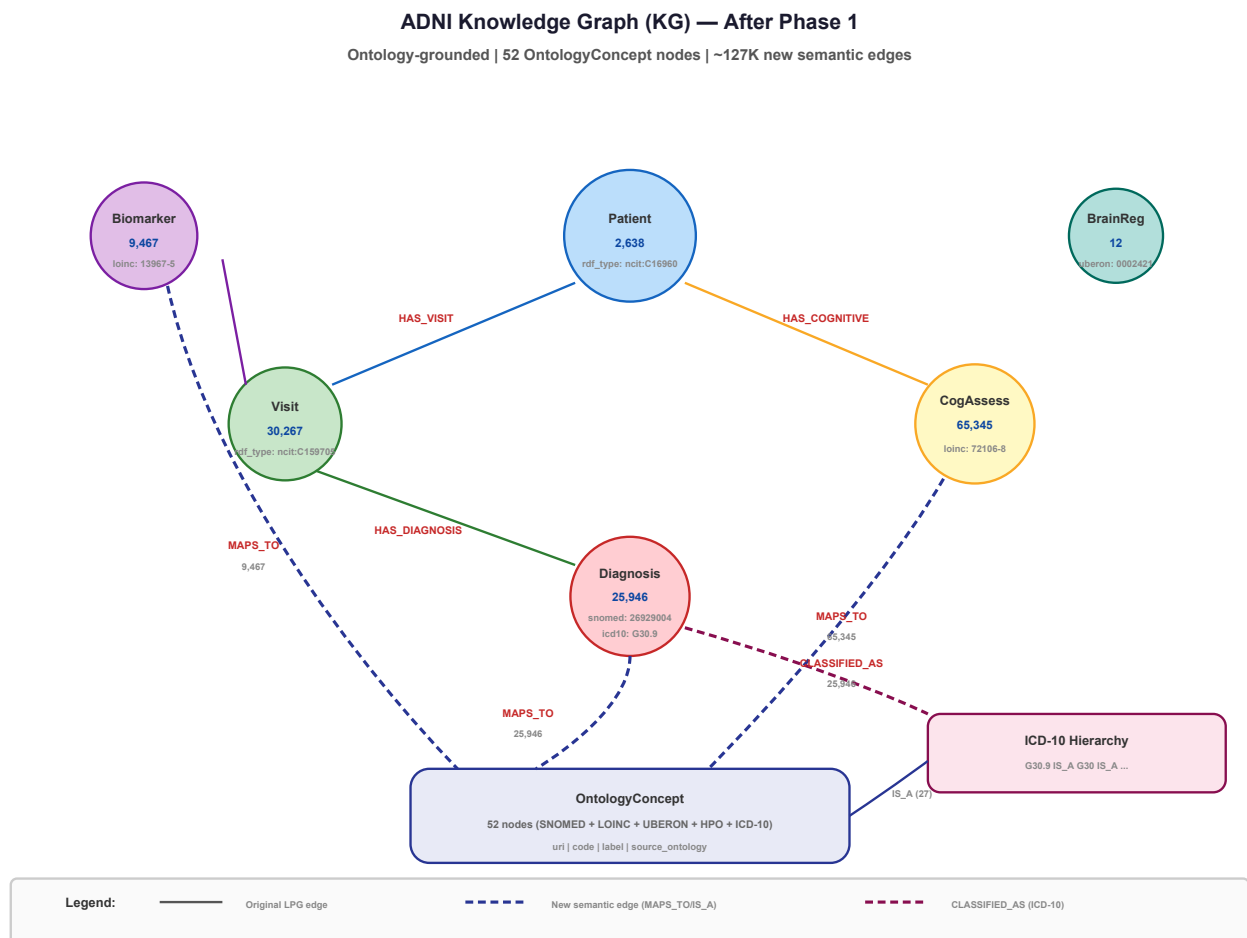


Figure 3: ADNI Knowledge Graph after Phase 1. Dashed lines show new semantic edges (MAPS_TO, CLASSIFIED_AS, IS_A). OntologyConcept nodes provide the semantic backbone.

6. Ontology Layer Detail

The ontology layer consists of 52 OntologyConcept nodes distributed across five ontology systems. Each concept has a globally unique URI (e.g., snomed:26929004), a human-readable label, a source_ontology identifier, and an optional code property.

Ontology	Concepts	IS_A Edges	MAPS_TO Source	MAPS_TO Count
SNOMED-CT	18	9	Diagnosis	25,946
LOINC	10	0	CogAssess + Biomarker	74,812
UBERON	14	13	BrainRegion	12
HPO	5	3	(future use)	0
ICD-10	5	2	Diagnosis (CLASSIFIED_AS)	25,946
TOTAL	52	27		126,716

Table 1: OntologyConcept distribution across ontology systems.

OntologyConcept Layer — IS_A Hierarchies MAPS_TO

52 concept nodes across 4 ontologies + ICD-10



Figure 4: OntologyConcept layer showing IS_A hierarchies within each ontology and MAPS_TO edge distribution.

7. Side-by-Side Comparison: LPG vs KG

The following table summarizes the key differences between the graph's LPG state (before Phase 1) and its KG state (after Phase 1):

Characteristic	LPG (Before)	KG (After Phase 1)
Data Model	Nodes + Edges + Properties	Nodes + Edges + Properties + OntologyConcepts
Node Semantics	App-specific labels (Patient, Visit, Diagnosis)	Ontology-grounded rdf_type = ncit:C16960
Relationship Semantics	String-typed edges	URI-annotated edges ro:RO_0000056
Disease Coding	diagnosis_code = 'AD'	snomed: 26929004 icd10: G30.9
Assessment Coding	test_name = 'MMSE'	loinc_code: 72106-8
Brain Region Coding	name = 'Hippocampus'	uberon: UBERON:0002421
Concept Taxonomy	NONE (flat)	IS_A hierarchies (27 edges)
Cross-Ontology Links	NONE	100,770 MAPS_TO 25,946 CLASSIFIED_AS
Machine Reasoning	Pattern matching	Ontology traversal + IS_A
External Interop.	Custom mappings req.	Standard URIs (AlzKB, FHIR)

Table 2: Feature-by-feature comparison of the LPG and KG states.

Labeled Property Graph vs Knowledge Graph

Key distinctions applied to the ADNI dataset

Property Graph	Knowledge Graph	Knowledge Graph
Data Model	Nodes + Edges + Properties	Nodes + Edges + Properties + Ontology Concepts
Node Semantics	Application-specific labels (Patient, Visit, Diagnosis)	Ontology-grounded labels rdf_type = nctit:C16960
Relationship Semantics	String-typed edges (HAS_VISIT, HAS_DIAGNOSIS)	URI-annotated edges uri = ro:RO_0000056
Disease Coding	diagnosis_code = 'AD' (local string)	snomed: 26929004 icd10: G30.9, mondo: 0004975
Assessment Coding	test_name = 'MMSE' (local string)	loinc_code: 72106-8 (global standard)
Brain Region Coding	name = 'Hippocampus' (local string)	uberon_code: UBERON:0002421 (global standard)
Concept Taxonomy	NONE (flat namespace)	IS_A hierarchies: SNOMED, UBERON, HPO, ICD-10 (27 edges)
Cross-Ontology Links	NONE	100,770 MAPS_TO + 25,946 CLASSIFIED_AS
External Interoperability	Requires custom mappings for each integration	Standard URIs enable direct AlzKB, FHIR linking
Machine Reasoning	Pattern matching only (Cypher queries)	Ontology traversal + IS_A inference + reasoning

Verdict: After Phase 1, the ADNI graph is a Knowledge Graph

100% ontology coverage | Formal hierarchies | Standard URIs | Cross-ontology linkage

Figure 5. Visual comparison table of LPG vs KG characteristics.

8. Classification Verdict

Based on the established definitions of Labeled Property Graphs and Knowledge Graphs, we classify the ADNI graph as follows:

KG Criterion	Status	Evidence
Ontology grounding on nodes	PASS	100% of Diagnosis, CogAssess, Biomarker, BrainRegion nodes
Formal relationship URIs	PASS	30 types, 1.2M rels with RO/SKOS/RDFS URIs
OntologyConcept layer	PASS	52 nodes across SNOMED/LOINC/UBERON/HPO/ICD-10
IS_A taxonomic hierarchies	PASS	27 IS_A edges in 4 ontology systems
MAPS_TO / CLASSIFIED_AS link	PASS	126,716 semantic edges linking data to concepts
rdf_type on nodes	PASS	Patient (ncit:C16960) Visit (ncit:C159705)
Global unique identifiers	PASS	URI property on all OntologyConcept nodes
Cross-ontology queryability	PASS	SNOMED-CT <-> ICD-10 via MAPS_TO + CLASSIFIED_AS

Table 3: Knowledge Graph criteria checklist — all criteria pass.

Conclusion: The ADNI Neo4j graph, after the completion of Phase 1 (Steps 17–20), satisfies **all standard criteria** for classification as a Knowledge Graph. It retains all LPG capabilities (fast traversal, property storage, Cypher queries) while adding the semantic layer that distinguishes a KG: ontology grounding, formal hierarchies, standard URIs, and cross-ontology linkage.

The graph is best described as an **LPG-based Knowledge Graph** or **Semantic Property Graph** — a hybrid architecture that combines the performance of Neo4j's LPG engine with the semantic richness of formal ontology systems.

9. What This Enables for Phases 2-4

The semantic upgrade completed in Phase 1 is not merely a classification exercise. It directly enables the subsequent research phases:

- **Phase 2 (Causal Discovery):** The ontology-grounded feature matrix (Step 21) can now use LOINC/SNOMED codes to standardize variable selection. Causal discovery algorithms (PC, FCI, GES) will produce edges that can be mapped back to `OntologyConcept` nodes via their standardized URIs, making discovered causal relationships interoperable.
- **Phase 3 (Validation):** The `MAPS_TO` and `CLASSIFIED_AS` edges enable direct alignment with AlzKB (118,902 entities). `SAME_AS` edges can be created between our `OntologyConcept` nodes and AlzKB entities using their shared SNOMED-CT/LOINC codes as join keys. This was impossible with application-specific strings.
- **Phase 4 (Defense Prep):** The `IS_A` hierarchies provide ready-made thesis figures. The semantic layer demonstrates the knowledge engineering contribution — the systematic methodology for transforming a clinical LPG into a semantically interoperable KG.

Graph Statistics After Phase 1

Metric	Count
Total nodes (original)	~407,000
OntologyConcept nodes (ICD-10)	5
OntologyConcept nodes (SNOMED/LOINC/UBERON/HPO)	47
Total OntologyConcept nodes	52
Original relationships	~1,160,000
CLASSIFIED_AS edges (Diagnosis -> ICD-10)	25,946
MAPS_TO edges (data -> ontology)	100,770
IS_A edges (hierarchy)	27
Total new semantic edges	126,743
Relationship types with URI property	30
Relationships with URI property	~1,200,000
Uniqueness constraints	12
Performance indexes	15

Table 4: Complete graph statistics after Phase 1 completion.

Report generated on February 24, 2026. Phase 1 (Steps 17–20) complete. Next: Phase 2 — Causal Discovery (Steps 21–23).